Table of Contents

..... 1
Experiment 7 : Image restoration using Mean Filters 1
Function for Airthmetic Mean filtering 1
Function for Geometric Mean filtering 1
Excercise 1 1
Conclusion 2
Excercise 2 3
Conclusion 4
Function for Harmonic Mean filtering 4
Excercise 3 4
Conclusion 5
Function for Contra Harmonic Mean filtering 6
Excercise 4 6
Conclusion 7
Excercise 5 8


```
% Yeshwantrao Chavan College of Engineering
% Department of Electronics and Telecommunication Engineering
% Session EVEN 2023-24
% Name of Student:Rutuja Saharkar
clc;
clear all;
close all;
```

Experiment 7 : Image restoration using Mean Filters

Function for Airthmetic Mean filtering

Filter Function function out = amfilter(I, m, n) a = floor((m - 1) / 2); b = floor((n - 1) / 2); F = padarray(I, [a, b], 'replicate', 'both'); [r, c] = size(F); out = zeros(r, c); for x = a + 1 : r - a for y = b + 1 : c - b for s = -a : a for t = -b : b out(x, y) = out(x, y) + F(x + s, y + t); end end out(x, y) = out(x, y) / (m * n); end end out = out(a + 1 : r - a, b + 1 : c - b); end

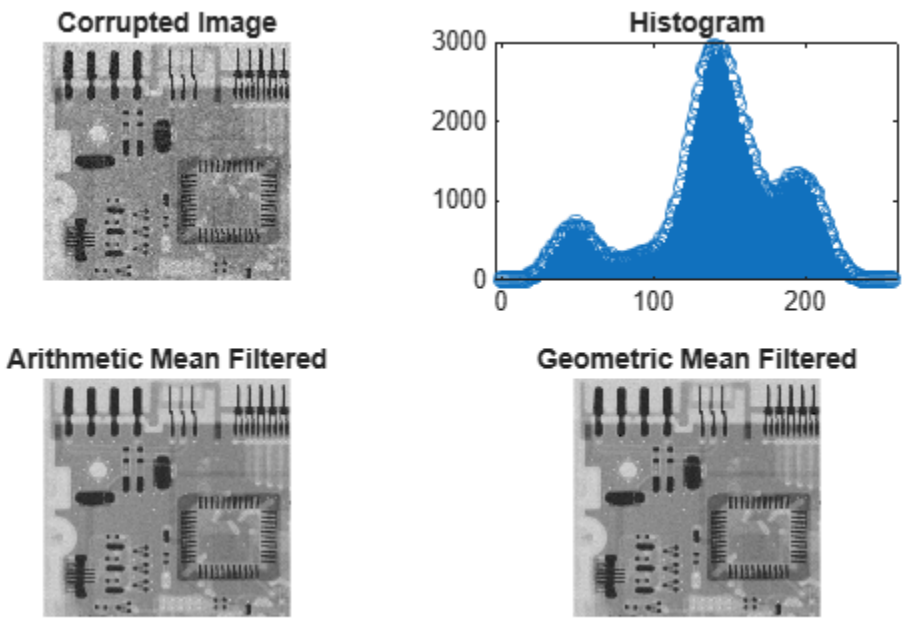
Function for Geometric Mean filtering

Filter Function function out = gmfilter(I,m,n) a = floor((m - 1) / 2); b = floor((n - 1) / 2); F = padarray(I, [a, b], 'replicate', 'both'); [r,c] = size(F); out = ones(r,c); for x = (a + 1):(r - a) for y = (b + 1):(c - b) product = 1; for s = -a:a for t = -b:b product = product * F(x + s, y + t); end end out(x, y) = product ^ (1 / (m * n)); end end out = out((a + 1):(r - a), (b + 1):(c - b)); end

Excercise 1

Read the image corrupted due to Gaussian Noise

```
Image = imread('Fig19_Gaussian.tif');
% Find its histogram
Hist1 = myhist(Image);
% Convert the image to double precision using im2double
IDouble = im2double(Image);
% Apply Arithmetic mean filter of size 3X3
arithmeticM = amfilter(IDouble,3,3);
% Apply Geometric mean filter of size 3X3
geometricM = gmfilter(IDouble, 3,3);
% Display the results in a 2x2 window
subplot(2,2,1),
imshow(Image),
title('Corrupted Image')
subplot(2,2,2)
stem(Hist1)
title("Histogram")
subplot(2,2,3),
imshow(arithmeticM),
title('Arithmetic Mean Filtered')
subplot(2,2,4),
imshow(geometricM),
title('Geometric Mean Filtered')
```

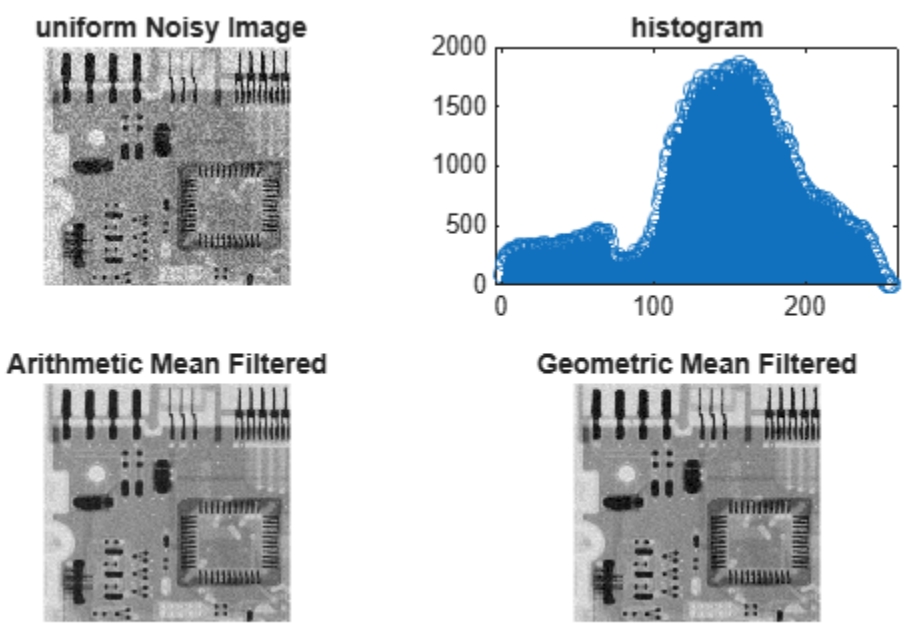


Conclusion

In this exercise we use arithmetic mean filter and geometric mean filter to eliminate Guassian Noise Present in a corrupted image(original image), An arithmetic mean filter operation on an image removes short tailed noise Gaussian type noise from the image at the cost of blurring the image. The arithmetic mean filter is defined as the average of all pixels within a local region of an image.The geometric mean filter is most widely used to filter out Gaussian noise. In general it will help smooth the image with less data loss than an arithmetic mean filter.

Excercise 2

```
%Perform image restoration using arithmetic and geometric mean filer on
% image corrupted due to Uniform noise
% Read the image corrupted due to Unifrom Noise
Image1 = imread('Fig20_Uniform.tif');
% Find its histogram
Hist2 = myhist(Image1);
% Convert the image to double precision using im2double
IDouble1 = im2double(Image1);
% Apply Airthmetic mean and geometric mean filter of size 3X3
arithmeticM = amfilter(IDouble1,3,3);
geometricM = gmfilter(IDouble1,3,3);
% Display the results in a 2x2 window
subplot(2,2,1)
imshow(Image1)
title(' uniform Noisy Image')
subplot(2,2,2)
stem(Hist2)
title("histogram")
subplot(2,2,3)
imshow(arithmeticM)
title('Arithmetic Mean Filtered')
subplot(2,2,4)
imshow(geometricM)
title('Geometric Mean Filtered')
```



Conclusion

In this exercise we use arithmetic mean filter and geometric mean filter to eliminate Uniform Noise Present in a corrupted image(original image), An arithmetic mean filter operation on an image removes short tailed noise Uniform type noise from the image at the cost of blurring the image. The arithmetic mean filter is defined as the average of all pixels within a local region of an image.The geometric mean filter is most widely used to filter out Uniform noise. In general it will help smooth the image with less data loss than an arithmetic mean filter.

Function for Harmonic Mean filtering

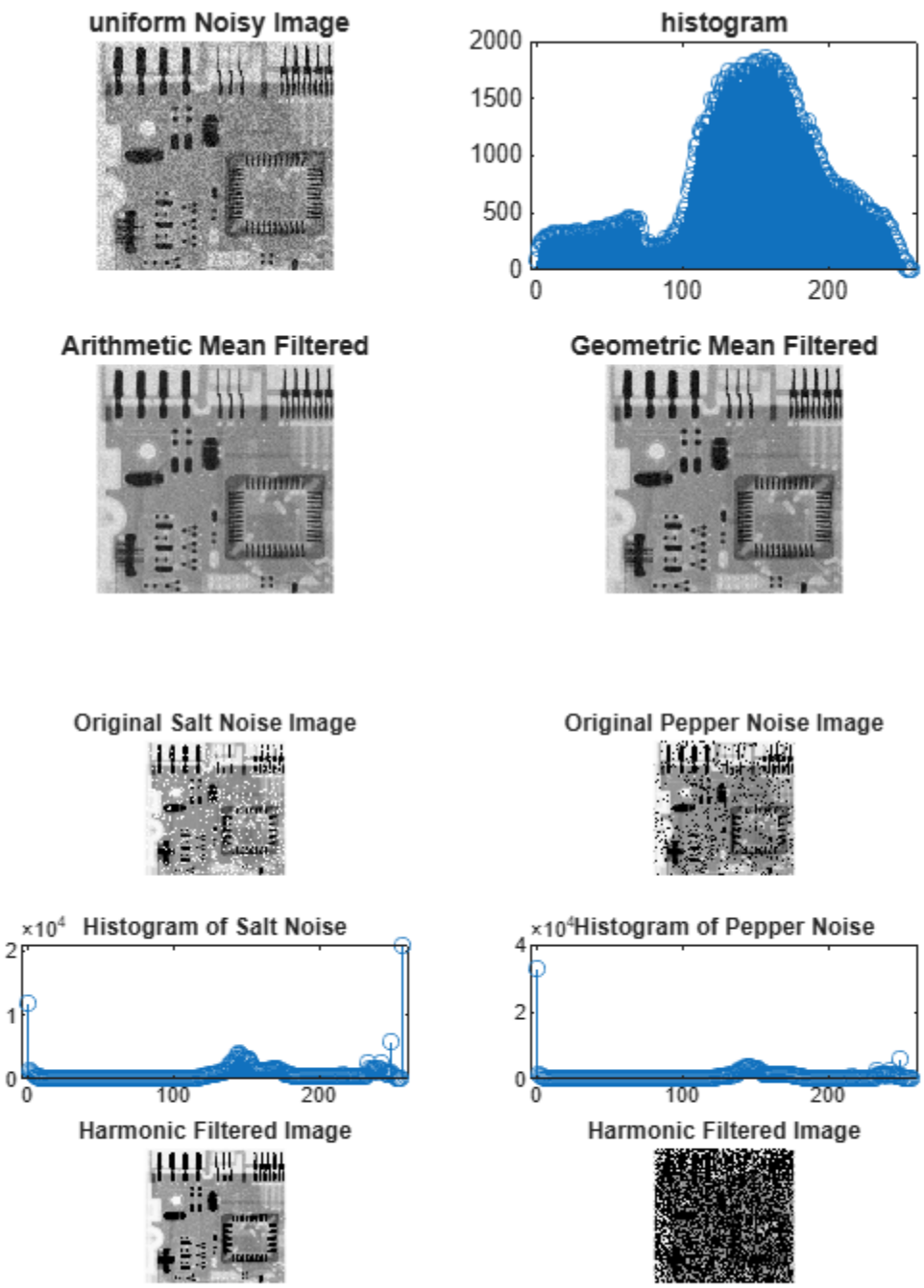
Filter Function (comment form) function out = hmfilter(I, m, n) a = (m - 1) / 2; b = (n - 1) / 2; F = padarray(I, [a, b], 'replicate', 'both'); [r,c] = size(F); out = zeros(r,c); for x = (a + 1):(r - a) for y = (b + 1):(c - b) for s = -a:a for t = -b:b out(x, y) = out(x, y) + (1 / F(x + s, y + t)); end end out(x, y) = (m * n) / out(x, y); end end out = out((a + 1):(r - a), (b + 1):(c - b)); end

Excercise 3

Perform image restoration using harmonic mean filer on image corrupted due Salt Noise and Pepper Noise Read the image corrupted due to Salt Noise

```
Image2 = imread('Fig21_Salt.tif');
% Find its histogram
Hist3 = myhist(Image2);
% Convert the image to double precision using im2double
IDouble2 = im2double(Image2);
% Apply harmonic mean filter of size 3X3
harmonicM1=hmfilter (IDouble2,3,3);
% Read the image corrupted due to Pepper Noise
Image3 = imread('Fig22_Pepper.tif');
% Find its histogram
Hist4 = myhist(Image3);
% Convert the image to double precision using im2double
IDouble3 = im2double(Image3);
% Apply harmonic mean filter of size 3X3
harmonicM2=hmfilter (IDouble3,3,3);
% Disply the results in  3 X 2 Window
figure,
subplot(3,2,1);
imshow(Image2);
title('Original Salt Noise Image');
subplot(3,2,2);
imshow(Image3);
title('Original Pepper Noise Image');
subplot(3,2,3);
stem(Hist3);
title('Histogram of Salt Noise');
subplot(3,2,4);
stem(Hist4);
title('Histogram of Pepper Noise');
subplot(3,2,5);
imshow(harmonicM1);
title('Harmonic Filtered Image');
```

```
subplot(3,2,6);
imshow(harmonicM2);
title('Harmonic Filtered Image');
```



Conclusion

The harmonic mean is a numerical average calculated by dividing the number of observations,or entries in the series,by the reciprocal of each number in the series.Thus,the harmonic mean is the reciprocal of the arithmetic

mean of the reciprocals.Harmonic mean filter is capable only to remove salt noise from an image.Pepper noise cannot be filtered using harmonic mean filter.

Function for Contra Harmonic Mean filtering

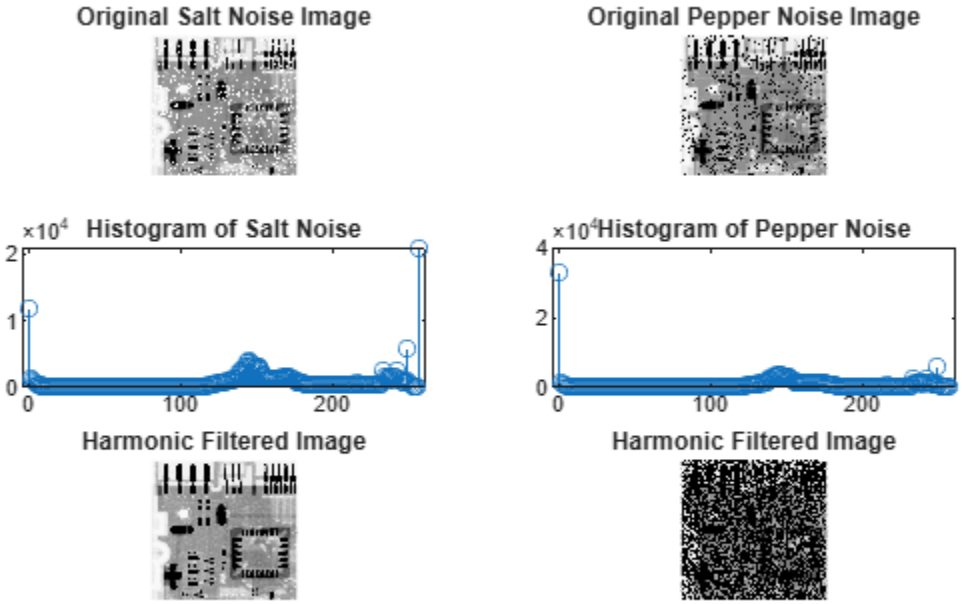
Filter Function (comment form) function out = chmfilter(I, m, n, Q) a = (m - 1) / 2; b = (n - 1) / 2; F = padarray(I, [a, b], 'replicate', 'both'); [r, c] = size(F); out = zeros(r, c); Nr = zeros(r, c); Dr = zeros(r, c); for x = (a + 1):(r - a) for y = (b + 1):(c - b) for s = -a:a for t = -b:b Nr(x, y) = Nr(x, y) + F(x + s, y + t) .^ (Q + 1); Dr(x, y) = Dr(x, y) + F(x + s, y + t) .^ Q; end end out(x, y) = Nr(x, y) / Dr(x, y); end end out = out((a + 1):(r - a), (b + 1):(c - b)); end

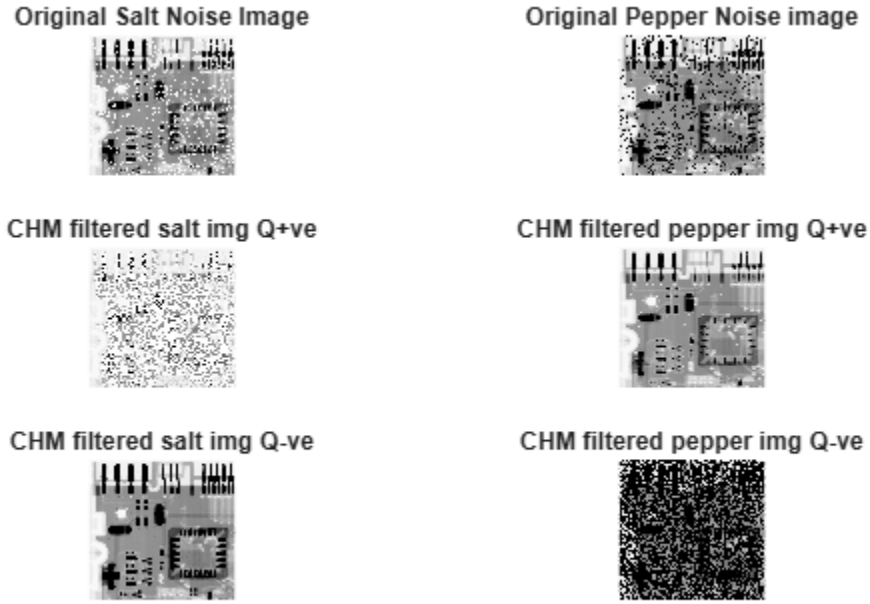
Excercise 4

Perform image restoration using contra harmonic mean filer on image corrupted due Salt Noise and Pepper Noise Read the image corrupted due to Salt Noise

```
Image4 = imread('Fig21_Salt.tif');
% Find its histogram
Hist5 = myhist(Image4);
% Convert the image to double precision using im2double
IDouble4 = im2double(Image4);
% Apply contra harmonic mean filter of size 3X3 with Q positive and Q
% negative
chms1=chmfilter(IDouble4,3,3,10);
chms2=chmfilter(IDouble4,3,3,-10);
% Read the image corrupted due to Pepper Noise
Image5 = imread('Fig22_Pepper.tif');
% Find its histogram
Hist6 = myhist(Image5);
% Convert the image to double precision using im2double
IDouble5 = im2double(Image5);
% Apply contra harmonic mean filter of size 3X3 with Q positive and Q
% negative
chms3=chmfilter(IDouble5,3,3,5);
chms4=chmfilter(IDouble5,3,3,-5);
% Disply the results
figure;
subplot(3,2,1);
imshow(Image4);
title('Original Salt Noise Image');
subplot(3,2,2);
imshow(Image5);
title('Original Pepper Noise image');
subplot(3,2,3);
imshow(chms1);
title('CHM filtered salt img Q+ve');
subplot(3,2,4);
imshow(chms3);
title('CHM filtered pepper img Q+ve');
subplot(3,2,5);
imshow(chms2);
title('CHM filtered salt img Q-ve');
subplot(3,2,6);
```

```
imshow(chms4);
title('CHM filtered pepper img Q-ve');
```





Conclusion

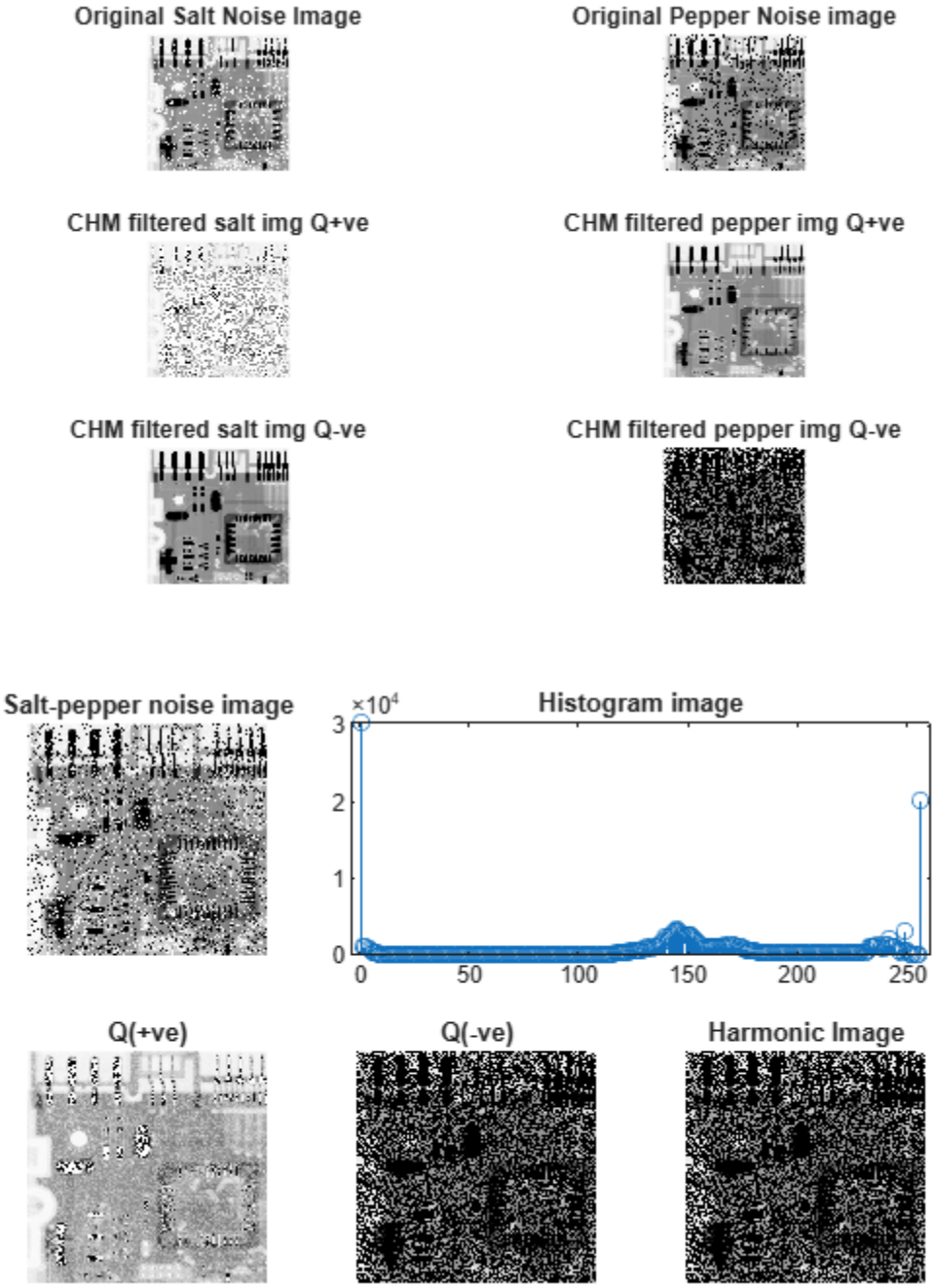
A contraharmonic mean filter reduces or virtually eliminates the effects of salt-and-pepper noise. For positive values of Q, the filter eliminates pepper noise. For negative values of Q it eliminates salt noise. It cannot do both

simultaneously.The contraharmonic filter is simply the arithmetic mean filter if Q = 0, and the harmonic mean filter if Q = −1. A larger region (filter size) yields a stronger filter effect with the drawback of some blurring.

Excercise 5

Perofrm harmonic and contraharmoic mean filtering (Q +Ve and -Ve) on image corrupted due to salt and pepper noise both.

```
I = imread('Fig23_Salt_Pepper.tif');
% Find its histogram
Hist7 = myhist(I);
% Convert the image to double precision using im2double
IDouble6 = im2double(I);
% Apply contra harmonic mean filter of size 3X3 with Q positive and Q
% negative
chms5=chmfilter(IDouble6,3,3,1);
chms6=chmfilter(IDouble6,3,3,-1);
% Apply harmonic mean filter of size 3X3
harmonicM3=hmfilter (IDouble6,3,3);
% Write your conclusion
figure,
subplot(2,3,1);
imshow(I)
title("Salt-pepper noise image")
subplot(2,3,[2,3]);
stem(Hist7)
title("Histogram image")
subplot(2,3,4);
imshow(chms5)
title("Q(+ve)")
subplot(2,3,5);
imshow(chms6)
title("Q(-ve)")
subplot(2,3,6);
imshow(harmonicM3)
title("Harmonic Image")
```



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